

Yiwei Liang

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Education

Massachusetts Institute of Technology

Sep 2024 – Present

Undergraduate student double major in Chemistry & Biology and Artificial Intelligence & Decision Making (GPA: 4.9/5.0)

Selected Coursework: Hallmarks of Cancer, Deep Learning (Grad), Human Physiology, AI and Decision Making in Medicine, Representation, Inference and Reasoning in AI, Organic Chemistry II, Fundamentals of Chemical Biology, Introduction to Computer Vision.

Princeton International School of Mathematics and Science

Sep 2021 – Jun 2024

The High School Affiliated to Renmin University of China

Sep 2017 – Jun 2021

Early Development Program

Honors and Awards

- **Regeneron Science Talent Search (STS) Scholar** Jan 2024
- **US National Chemistry Olympiad Study Camp (Top 20 Nationwide)** Jun 2023
- **S.-T. Yau High School Science Award (USA Regional Silver Medal)** Nov 2023

Publications

- **ProtQueSt: Query-Conditioned Retrieval-Augmented Generation for Protein Function Annotation** May 2026
L. Ma*, Y. Liang*, Y. Yu**, C. J. Qi**
The 2026 Workshop on Generative and Agentic AI for Biology, ICML 2026. **Spotlight paper.**
- **OncRef: A Pan-Cancer PRISM Reference of Cancer Therapeutics** May 2026
M. G. Rees, M. J. Emmett, ..., Y. Liang, ..., J. A. Roth, W. R. Sellers (11 of 24 authors)
In review at Cell.

Research Experience

- **Cancer Genomics Research** Sep 2024 – Present
William Sellers Lab, Broad Institute of MIT and Harvard
Mentor: Matthew J. Emmett, M.D., Ph.D.
Research on *in vivo* cancer dependencies utilizing a Cas9/Anti-CRISPR switch in the tumor microenvironment. Paper in review at Cell.
- **Cancer Immunotherapy Research** May 2022 – Aug 2023
Moustafa Gabr Lab, Weill Cornell Medicine
Identified and validated a small-molecule TIGIT inhibitor for immune checkpoint therapy: performed virtual screening and confirmed binding/activity via cell-free and cell-based assays.

Projects

- **Representation Efficiency in Neural Networks: Evidence in Multilingual Mathematical Benchmarks** Oct 2025 – Dec 2025
Link to project: <https://github.com/bowenyu066/language-shapes-reasoning>.
Investigated how representation choice affects neural reasoning efficiency in Transformers by multilingual mathematical benchmarks. Built a comprehensive evaluation framework testing SOTA models and 8B-parameter open-source models on GSM8K and MMATH across multiple languages. Discovered that well-trained models achieve representation-invariant reasoning, and when tokenizers align with denser representations, 510% token efficiency gains can be achieved. Demonstrated

that representation bottlenecks stem from training distribution rather than architecture by reducing English-Chinese performance gap through LoRA fine-tuning. Developed modular Python infrastructure featuring multitokenizer analysis pipeline and automated evaluation across API and local models.

Leadership and Service

- **President**, SYMPH Undergraduate Student Association *Sep 2025 – Present*
The Stanford–Yale–MIT–Princeton–Harvard (SYMPH) Undergraduate Student Association is a Chinese student organization with **500+** members. Organized Thanksgiving and summer multi-day retreats for **300+** participants.
- **President**, MIT Greater China Student Network *Sep 2025 – Present*
Organized multiple events for **100+** MIT undergraduate Chinese students.
- **Publicity Chair**, MIT Global Health Alliance *Jan 2026 – Present*
Lead publicity and communications efforts for campus events and initiatives related to global health.
- **Residential Director**, MedLinks *Oct 2025 – Present*
Support student health and wellness within the residential community by serving as a peer health resource and coordinator. Hosted residential events and provided resources for those in need.
- **Scientific Committee Chair**, Harvard–MIT Chemistry Olympiad (HMChO) *Mar 2025 – Present*
Lead the problem-writing and grading teams; oversee quality control for the annual competition with **100+** high school participants globally.
- **Board Member**, MIT Science Bowl *May 2024 – Present*
Author for the chemistry section; help organize the annual national tournament with **30+** high school teams.

Volunteer Experience

- **Emergency Radiology Volunteer**, Massachusetts General Hospital *Oct 2024 – Present*
Assist with patient transport and imaging workflows in a fast-paced clinical environment; support patients of all ages through diagnostic pathways. Total 200+ hours of service. SSSS
- **Volunteer**, American Chemical Society *2022 – 2024*
Taught free online USNCO preparation sessions to **200+** students; wrote annotated solutions and co-organized a webinar for **500+** attendees.